

Automating TIL Evaluation in Breast Cancer

Accurate prognosis is essential for effective breast cancer treatment. One of the most important biomarkers for this purpose is **Tumor-Infiltrating Lymphocytes (TILs)**—immune cells that migrate into tumor tissue, indicating the body's immune response against cancer. Higher TIL levels are linked to better outcomes, especially in **Triple-Negative** and **HER2-Positive** breast cancers.

Currently, TIL assessment is performed manually by pathologists through visual inspection of histopathology slides. This process is **subjective**, **time-consuming**, and **difficult to scale**, as results may vary between experts and require extensive effort.

To overcome these limitations, this project focuses on **automated lymphocyte detection** using deep learning. The dataset is sourced from the **TIGER Challenge** and consists of high-resolution H&E-stained whole slide image patches with bounding box annotations for lymphocytes, enabling object detection tasks.

The proposed approach uses **Faster R-CNN with a ResNet-50 backbone**, selected for its strong performance in detecting small objects within complex images. **Transfer learning** is applied by fine-tuning a pre-trained ResNet-50 model to identify lymphocytes accurately.

Key challenges include the **small size of lymphocytes**, their **visual similarity to other cell nuclei**, and **high cell density** in inflamed regions. Despite these challenges, automated TIL detection offers a consistent, scalable, and unbiased alternative to manual scoring, with the potential to improve clinical decision-making and patient outcomes.